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November 11, 2021

Attn: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

10 CFR 50.73

SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-387/2021-003-01
UNIT 1 LICENSE NO. NPF-14
PLA-7971

Docket No. 50-387

Attached is Licensee Event Report (LER) 50-387/2021-003-01. The LER supplement reports an event involving an automatic scram due to a main turbine trip. The condition is being reported in accordance with 10 CFR 50.73(a)(2)(iv)(A) as an event that resulted in an automatic actuation of the Reactor Protection System (including a reactor scram).

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new or revised regulatory commitments.

A handwritten signature in black ink, appearing to read "K. Cimorelli", written over a horizontal line.

K. Cimorelli

Attachment: LER 50-387/2021-003-01

Copy: NRC Region I
Ms. A. Klett, NRC Project Manager
Mr. C. Highley, NRC Senior Resident Inspector
Mr. M. Shields, PA DEP/BRP



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form <https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk ail: aira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Susquehanna Steam Electric Station Unit 1					2. Docket Number 05000387					3. Page 1 of 3					
4. Title Automatic Reactor Scram Due to Main Turbine Trip Caused by a 'C' Isophase Bus Ground Fault															
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved						
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name			Docket Number			
07	21	2021	2021	- 003 -	01	11	11	2021	Facility Name			Docket Number			
9. Operating Mode					10. Power Level										
1					100										
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)															
10 CFR Part 20		☐ 20.2203(a)(2)(vi)			☐ 50.36(c)(2)			☒ 50.73(a)(2)(iv)(A)			☐ 50.73(a)(2)(x)				
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)			☐ 50.46(a)(3)(ii)			☐ 50.73(a)(2)(v)(A)			10 CFR Part 73				
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)			☐ 50.69(g)			☐ 50.73(a)(2)(v)(B)			☐ 73.71(a)(4)				
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)			☐ 50.73(a)(2)(i)(A)			☐ 50.73(a)(2)(v)(C)			☐ 73.71(a)(5)				
☐ 20.2203(a)(2)(i)		10 CFR Part 21			☐ 50.73(a)(2)(i)(B)			☐ 50.73(a)(2)(v)(D)			☐ 73.77(a)(1)(i)				
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)			☐ 50.73(a)(2)(i)(C)			☐ 50.73(a)(2)(vii)			☐ 73.77(a)(2)(i)				
☐ 20.2203(a)(2)(iii)		10 CFR Part 50			☐ 50.73(a)(2)(ii)(A)			☐ 50.73(a)(2)(viii)(A)			☐ 73.77(a)(2)(ii)				
☐ 20.2203(a)(2)(iv)		☐ 50.36(c)(1)(i)(A)			☐ 50.73(a)(2)(ii)(B)			☐ 50.73(a)(2)(viii)(B)							
☐ 20.2203(a)(2)(v)		☐ 50.36(c)(1)(ii)(A)			☐ 50.73(a)(2)(iii)			☐ 50.73(a)(2)(ix)(A)							
☐ Other (Specify here, in Abstract, or in NRC 366A).															
12. Licensee Contact for this LER															
Licensee Contact D. R. Smith, Senior Engineer – Nuclear Regulatory Affairs										Phone Number (Include Area Code) 570-542-1377					
13. Complete One Line for each Component Failure Described in this Report															
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS						
A	EL	IPBU	G080	Y											
14. Supplemental Report Expected										15. Expected Submission Date			Month	Day	Year
☒ No		☐ Yes (If yes, complete 15. Expected Submission Date)													
16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)															
On July 21, 2021, at approximately 18:26, Susquehanna Steam Electric Station Unit 1 reactor automatically scrambled due to a main turbine trip. Both divisions of the Reactor Protection System (RPS) actuated and all control rods inserted. This event was reported by Event Notification 55370 in accordance with 10 CFR 50.72(b)(2)(iv)(B) and (b)(3)(iv)(A). This event is also reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) as an event that resulted in automatic actuation of a system listed in 10 CFR 50.73(a)(2)(iv)(B). The cause of the event was a dislodged de-ionizing baffle plate that contacted the 'C' isophase bus causing a ground fault. The root cause was determined to be accelerated wear of the de-ionizing baffle tack welds due to increased isophase bus forced air. Corrective actions included replacement and tack welding the dislodged and loose baffle plates and implementing an Engineering Change to modify the Unit 1 Main Generator-end 'A' and 'C' isophase bus de-ionizer baffle plates to install an aluminum retaining rod to secure the baffle plates together. There were no actual consequences to the health and safety of the public as a result of this event.															

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME

Susquehanna Steam Electric Station Unit 1

2. DOCKET NUMBER

05000-387

3. LER NUMBER

YEAR	SEQUENTIAL NUMBER	REV NO.
2021	- 003 -	01

NARRATIVE**CONDITIONS PRIOR TO EVENT**

Unit 1 – Mode 1, approximately 100 percent Rated Thermal Power

Unit 2 – Mode 1, approximately 100 percent Rated Thermal Power

There were no structures, systems, or components that were inoperable at the start of the event that contributed to the event.

EVENT DESCRIPTION

On July 21, 2021, at approximately 18:26, Susquehanna Steam Electric Station Unit 1 reactor automatically scrammed due to a main turbine [EIS System/Component Code: TA/TRB] trip. The Unit 1 Control Room received indication of a main turbine trip with both divisions of the Reactor Protection System (RPS) [EIS System Code: JC] actuated and all control rods inserted. The Reactor Recirculation Pumps [EIS System/Component Code: AD/P] tripped on End of Cycle Recirculation Pump Trip. Operations subsequently maintained reactor water level at the normal operating band using the Reactor Feedwater system [EIS System Code: SJ].

This event was reported by Event Notification 55370 in accordance with 10 CFR 50.72(b)(2)(iv)(B) and (b)(3)(iv)(A). This event is also reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) as an event that resulted in automatic actuation of a system listed in 10 CFR 50.73(a)(2)(iv)(B).

CAUSE OF EVENT

The cause of the event was a dislodged de-ionizing baffle plate that contacted the 'C' isophase bus [EIS System/Component Code: EL/IPBU] causing a ground fault. The root cause was determined to be accelerated wear of the de-ionizing baffle tack welds due to increased isophase bus forced air.

Contributing causal factors included: 1) industry isophase bus events were not effectively reviewed to identify and implement station learnings for prevention of isophase bus failures and 2) visual inspection guidelines used to inspect the isophase de-ionizer grids for fatigue were inadequate.

ANALYSIS/SAFETY SIGNIFICANCE

The scram did not require or result in the actuation of any Emergency Core Cooling System or the Reactor Core Isolation Cooling system [EIS System Code: BN] and no main steam relief valves [EIS System/ Component Code: SB/RV] opened. All safety systems responded properly during the event. There were no actual consequences to the health and safety of the public as a result of this event.

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2021

SEQUENTIAL

NUMBER

- 003 -

REV

NO.

01

NARRATIVE**CORRECTIVE ACTIONS**

Key corrective actions included the following:

1. Replaced and tack welded the dislodged and loose baffle plates.
2. Implemented an Engineering Change to modify the Unit 1 Main Generator-end 'A' and 'C' isophase bus de-ionizer baffle plates to install an aluminum retaining rod to secure the baffle plates together.

Actions will be taken to address the extent of condition on Unit 2 as required.

COMPONENT FAILURE INFORMATION

Component Identification – 1G105

Component Name – Generator Bus, Isolated Phase Bus and Auxiliary

Manufacturer – General Electric

PREVIOUS OCCURRENCES

None.